

Module Two Assignment

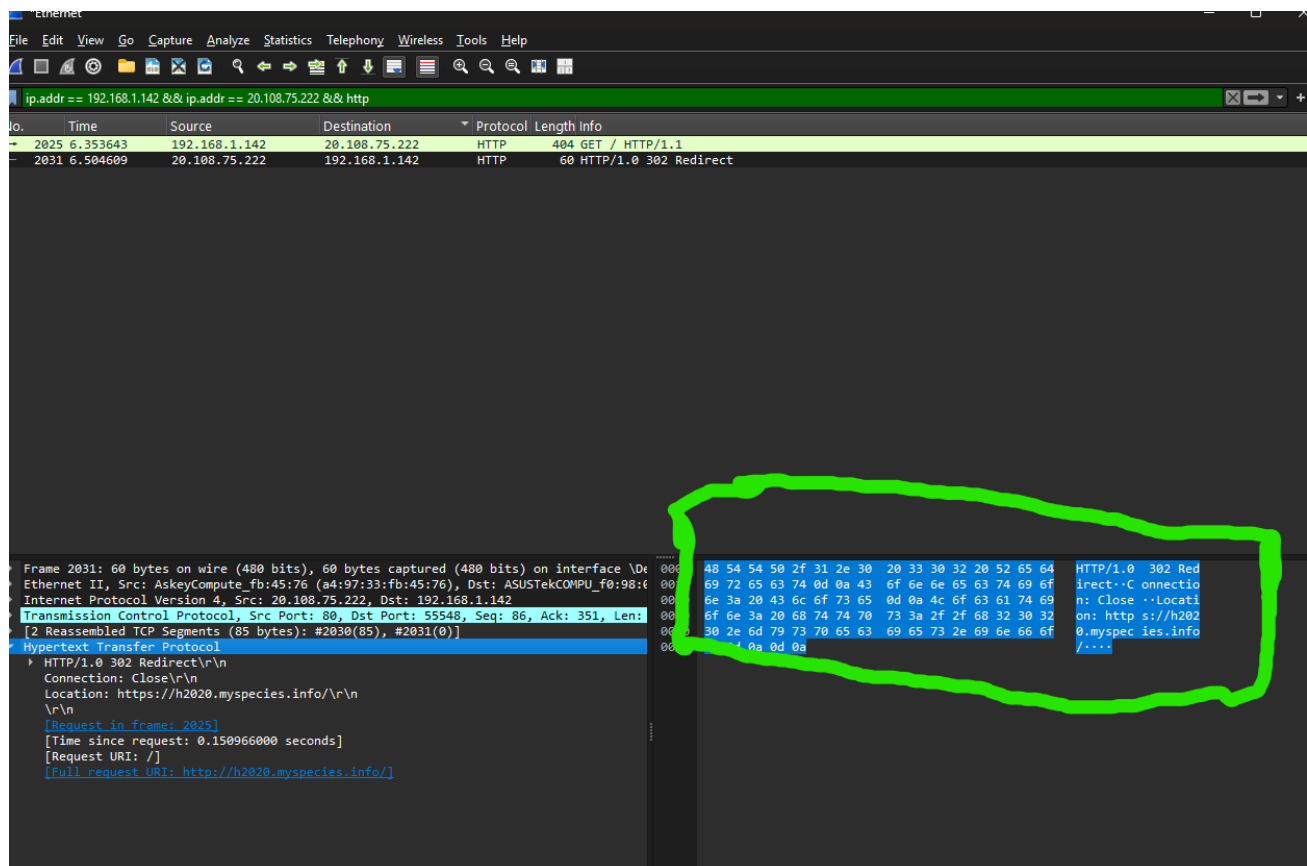
Introduction to Wireshark & Deep Packet Inspection

Part 1: Capture Traffic

Wireshark packet capture interface showing a list of network packets. The interface includes a menu bar, toolbar, and a packet list pane. The packet list pane shows a series of packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. Handwritten annotations in red and blue circles highlight specific protocols (TCP, HTTP, TLSv1.3) and a status 'O = DOWN'.

No.	Time	Source	Destination	Protocol	Length	Info
2000	6.202170	192.168.1.142	20.108.75.222	TCP	66	55548 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2023	6.353464	20.108.75.222	192.168.1.142	TCP	66	80 → 55548 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
2024	6.353514	192.168.1.142	20.108.75.222	TCP	54	55548 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2025	6.353643	192.168.1.142	20.108.75.222	HTTP	404	GET / HTTP/1.1
2029	6.504570	20.108.75.222	192.168.1.142	TCP	60	80 → 55548 [ACK] Seq=1 Ack=351 Win=64128 Len=0
2030	6.504570	20.108.75.222	192.168.1.142	TCP	139	80 → 55548 [PSH, ACK] Seq=1 Ack=351 Win=64128 Len=85 [TCP PDU reassembled in 2031]
2031	6.504609	20.108.75.222	192.168.1.142	HTTP	60	HTTP/1.0 302 Redirect
2032	6.504621	192.168.1.142	20.108.75.222	TCP	54	55548 → 80 [ACK] Seq=351 Ack=87 Win=65280 Len=0
2033	6.504742	192.168.1.142	20.108.75.222	TCP	54	55548 → 80 [FIN, ACK] Seq=351 Ack=87 Win=65280 Len=0
2034	6.531526	192.168.1.142	20.108.75.222	TCP	66	55549 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2042	6.653092	20.108.75.222	192.168.1.142	TCP	60	80 → 55548 [ACK] Seq=87 Ack=352 Win=64128 Len=0
2043	6.686131	20.108.75.222	192.168.1.142	TCP	66	443 → 55549 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
2044	6.686195	192.168.1.142	20.108.75.222	TCP	54	55549 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
2045	6.686576	192.168.1.142	20.108.75.222	TLSv1.3	1959	Client Hello (SNI=h2020.myspecies.info)

Part 2: Digging into Packets



How a malicious attacker might use a passive packet capturing tool like Wireshark to violate the CIA+authenticity+accountability pentagon.

A passive packet capturing tool like Wireshark can give malicious attackers deep visibility into network traffic, which can create opportunities to violate every aspect of the CIA+Authenticity+Accountability pentagon. By observing unencrypted packets an attacker could harvest credentials and later reuse them, which would undermine authenticity because the server would accept forged requests as if they came from the real user. This type of impersonation attack allows the attacker to effectively “become” the user without ever needing to actively break into the system. With stolen sessions or passwords, an attack could also modify or inject data which would violate integrity. Even though passive sniffing does not directly bring down systems or reduce availability, it equips malicious attackers with valuable reconnaissance such as services, versions, and usage patterns. These types of information can later be used for a targeted denial of service (DOS) attack or injection attempts. Observing traffic through a tool like Wireshark demonstrates how unprotected communication can compromise confidentiality, integrity, authenticity, and accountability. Which really highlights the importance of defense in depth.

Resources